

Preliminaries

Let $\mathcal{D}_d$ be the n -dimensional quantum states

$$\mathcal{D}_d = \{\rho \geq 0 : \rho \in M_d(\mathbb{C}), \text{Tr}(\rho) = 1\}$$

and consider the seperable states

$$\mathcal{SEP}_{d_1, d_2} = \left\{ \sum_i p_i \rho_i \otimes \sigma_i : \rho_i \in \mathcal{D}_{d_1}, \sigma_i \in \mathcal{D}_{d_2}, \sum_i p_i = 1 \right\}$$

It is apparent that $\mathcal{SEP}_{d_1, d_2} \subset \mathcal{D}_{d_1 d_2}$. One useful criterion for classifying seperable states is called Peres–Horodecki criterion, or PPT (positive partial transpose) criterion. It states that if a $X \in \mathcal{SEP}_{d_1, d_2}$ then $(\text{id} \otimes \top)(X) \geq 0$. In other words, consider

$$\mathcal{PPT}_{d_1, d_2} = \{X \in \mathcal{D}_{d_1 d_2} : (\text{id} \otimes \top)(X) \geq 0\}$$

Remark 1. In fact, this definition can be extend to any semi positive definite matrix, as

$$\mathcal{SEP}_{d_1, d_2} = \left\{ \sum_i X_i \otimes Y_i : X_i \in M_{d_1}, Y_i \in M_{d_2}, X_i, Y_i \geq 0 \right\}$$

and

$$\mathcal{PPT}_{d_1, d_2} = \{X \geq 0 : (\text{id} \otimes \top)(X) \geq 0\}$$

This is because $(\text{id} \otimes \top)(X) = \sum_i X_i \otimes Y_i^\top \geq 0$ for $X \in \mathcal{SEP}_{d_1, d_2}$, thus we have $\mathcal{SEP}_{d_1, d_2} \subset \mathcal{PPT}_{d_1, d_2} \subset \mathcal{D}_{d_1 d_2}$

Now we will introduce Schmidt number to charaterize the seperable states.

Definition 1. Consider the one rank matrix $X = xx^* \in M_{d_1} \otimes M_{d_2}$. Then, define the Schmidt number of this matrix as

$$\mathcal{SN}(X) = \text{rank}(\text{id}_{d_1} \otimes \text{tr}_{d_2})(X)$$

Remark 2. For general positive semi definite matrix $X \in M_{d_1} \otimes M_{d_2}$, we can define its Schmidt number as its convex roof. That is,

$$\mathcal{SN}(X) = \min \left\{ r : X = \sum_{i=1}^m x_i x_i^*, \text{ with } \mathcal{SN}(x_i x_i^*) \leq r \right\}$$

Remark 3. We can show that $\mathcal{SN}(X) = 1$ if and only if X is seperable.

Proof. Lets consider the one rank case. Since $\mathbb{C}^{d_1} \otimes \mathbb{C}^{d_2} \cong M_{d_1 \times d_2}(\mathbb{C})$, by

$$\sum_{i,j} x_{ij} u_i \otimes v_j \mapsto X = [x_{ij}] = \sum_{ij} x_{ij} u_i v_j^*$$

Next, observe $(\text{id} \otimes \text{tr})(xx^*) = XX^*$ and $\text{rank}(XX^*) = \text{rank}(X)$, then it follows that $\text{rank}(X) = 1$ if and only if $x = u \otimes v$. In general, for $X \in M_{d_1} \otimes M_{d_2}$, $X \geq 0$ and $\mathcal{SN}(X) = 1$, we have $X = \sum_i x_i x_i^*$ with $\mathcal{SN}(x_i x_i^*) = 1$. Then, X is of the form $X = \sum_i (a_i a_i^*) \otimes (b_i b_i^*) \in \mathcal{SEP}_{d_1, d_2}$. Conversely, let $X = \sum_i A_i \otimes B_i$ with $A_i, B_i \geq 0$. Consider $A = \sum_j \lambda_j a_j a_j^*$ and $B = \sum_k \tilde{\lambda}_k b_k b_k^*$. Then, $X = \sum_{i,j,k} \left(\sqrt{\lambda_j \tilde{\lambda}_k} a_i \otimes b_k \right) \left(\sqrt{\lambda_j \tilde{\lambda}_k} a_i \otimes b_k \right)^*$. This demonstrate that $\mathcal{SN}(X) = 1$. $\square$

This classifies the seperable from entanglement. However, we are unable to check the Schmidt number of matrices most of the time, since solving its Schmidt number itself is a optimization problem, which is an NP-hard problem.

It is interesting to ask how Random matrices with specific Schmidt number act in asymptotic sense. To be more specific, consider a Random matrices X_d that converges to μ . Then, if its Schmidt number converges $\mathcal{SN}(X_d) \rightarrow \mathcal{SN}(x)$. Can we find a Hermitian preserving map φ satisfying UC condition such that μ^φ preserves some $\mathcal{SN}$ charateristic. That is, what is the relations of

$$\mathcal{SN}(X_d) \rightarrow f(\mu^\varphi)$$

in asymptotic sense.

Schmidt number

Now, we will present the meaning of Schmidt number. In this section, we are basically chasing the following diagram.

$$\begin{array}{ccc} M_m \otimes M_k & \xrightarrow{\text{id}_m \otimes \iota} & M_m \otimes M_d \\ \varphi \otimes \text{id}_k \downarrow & & \downarrow \varphi \otimes \text{id}_d \\ M_n \otimes M_k & \xrightarrow{\text{id}_n \otimes \iota} & M_n \otimes M_d \\ \text{id}_n \otimes \text{Tr}_k \downarrow & \swarrow \text{id}_n \otimes \text{Tr}_d & \\ M_n & & \end{array}$$

where we have $\{E_{ij}\}$ be the basis of M_m , $\{F_{kl}\}$ be the basis of M_n , $\{H_{\alpha\beta}\}$ be the basis of M_d , and $\{G_{pq}\}$ be the basis of M_k .

Remark 4. The graph shown above commutes with the inclusion map given by ι

$$\iota(G_{pq}) = BG_{pq}B^* = H_{pq}$$

Also, note that in the following context, we always denote $k \leq \min(n, m, d)$. Let's first see some results established [1].

Lemma 5. Let P' be a one rank projection in $M_m \otimes M_d$. Then, there is a inclusion map $\iota : M_m \otimes M_k \rightarrow M_m \otimes M_d$ such that $\iota(P) = P'$ for some projection $P \in M_m \otimes M_k$ if and only if $\mathcal{SN}(P') \leq k$.

Proof. For any one rank orthogonal projection $P' = vv^*$ for some $v = \sum_{i=1}^m \sum_{j=1}^d v_{ij} e_i \otimes h_j$. If there is some $P = pp^*$ for some $p = \sum_{i=1}^m \sum_{j=1}^k p_{ij} e_i \otimes g_j$ such that $\iota(P) = vv^*$, it then follows that $\text{id} \otimes \text{tr}(vv^*) = WW^*$ for $W = \sum_{i=1}^m \sum_{j=1}^k p_{ij} e_i g_j^*$. Since $\text{rank}(WW^*) = \text{rank}(W) \leq k$, we have $\mathcal{SN}(P') \leq k$.

Conversely, let vv^* be one rank projection with $\mathcal{SN}(vv^*) \leq k$ for some $v = \sum_{i=1}^m \sum_{j=1}^d v_{ij} e_i \otimes h_j$. This means that for $W = \sum_{i=1}^m \sum_{j=1}^d v_{ij} e_i h_j^*$, $\text{rank}(W) = \text{rank}(WW^*) \leq k$. Thus, there are k linearly independent vectors $\{\tilde{h}_1, \dots, \tilde{h}_k\}$ that forms a basis of $\text{im}(W^*)$. Consider the inclusion maps $\iota : M_m \otimes M_k \rightarrow M_m \otimes M_d$ by

$$E_{ij} \otimes G_{\alpha\beta} \mapsto E_{ij} \otimes \tilde{H}_{\alpha\beta}$$

By construction, vv^* lies in the $\text{im}(\iota)$ meaning there exist one rank projection P such that $\iota(P) = P'$. The result follows. $\square$

Corollary 6. Let $\varphi : M_m \rightarrow M_n$ be a Hermitian preserving maps. Then, φ is k -positive if and only if

$$\varphi \otimes \text{id}_d(vv^*) \geq 0$$

for $\mathcal{SN}(vv^*) \leq k \leq \min(n, m, d)$ and $vv^* \in M_m \otimes M_d$

Proof. Suppose $\mathcal{SN}(vv^*) \leq k$, from Lemma 5, we have $pp^* \in M_n \otimes M_k$ and an inclusion map such that $\iota(pp^*) = vv^*$. Then, it follows that $\varphi \otimes \text{id}_d(vv^*) = \iota(\varphi \otimes \text{id}_k(pp^*)) \geq 0$. Conversely, for any positive semi-definite matrix X , it admits one rank decomposition. Thus, we only check the one rank case. Then, $\varphi \otimes \text{id}_k(xx^*) = \varphi \otimes \text{id}_d(vv^*) \geq 0$ for some $\iota(xx^*) = vv^* \in M_m \otimes M_d$. $\square$

Remark 7. Note that $v \in M_m \otimes M_d$ with $\mathcal{SN}(vv^*) \leq k \leq \min(m, d)$. We have $v = \sum_{i=1}^m \sum_{j=1}^d v_{ij} e_i \otimes h_j = (\text{id}_m \otimes B)O$, where $Be_i = \sum_{j=1}^d v_{ij} h_j$ with $\text{rank}(B) \leq k$ and $O = \sum_i e_i \otimes e_i$. Then, the above condition is equivalent to

$$\varphi \otimes \text{id}_d((\text{id}_m \otimes B)OO^*(\text{id}_m \otimes B)^*) = (\text{id}_n \otimes B)C_\varphi(\text{id}_n \otimes B)^* \geq 0$$

This is equivalent of saying $l^* C_\varphi l \geq 0$ for all v with $\mathcal{SN}(ll^*) \leq k$

Theorem 8. Let X be a positive semi-definite matrix on $M_m \otimes M_d$. Then $\mathcal{SN}(X) \geq k + 1$ if and only if there exist a k -positive linear map $\varphi : M_m \rightarrow M_n$ with $1 \leq k \leq \min(m, d)$ such that

$$(\varphi \otimes \text{id}_d)(X) \not\geq 0$$

Proof. Let $X \in M_m \otimes M_d$ be a PSD matrix with $\mathcal{SN}(X) \leq k$. Then, X admits some one rank decomposition with each projection has Schmidt number smaller than k . According to Corollary 6, we have $\varphi \otimes \text{id}_d(X) \geq 0$.

Conversely, since the positive semi-definite matrix with Schmidt number less than k forms a convex set. According to Hahn Banach separation theorem, there exist a functional f separates them. Now, according to Riesz representation theorem, we can find a matrix $M \in M_m \otimes M_d$ such that $f(X) = \text{Tr}(W^*X)$. Based on W , we can use Choi isomorphism to construct our desired k -positive map. $\square$

Using the Theorem 8, we can redefine our Schmidt number through k -positive map.

Definition 2. Given a positive semi-definite matrix $X \in M_m \otimes M_d$. Its Schmidt number is given by

$$\mathcal{SN}(X) = \max\{k : \exists k\text{-positive map } \varphi \text{ s.t. } \varphi \otimes \text{id}_d(X) \not\geq 0\} + 1$$

This means that in order to find the Schmidt number of a Random matrix. We can find the maximum number of k such that there is some k -positive map φ that makes $\varphi \otimes \text{id}_d(X_d) \not\geq 0$.

Recently, it has been shown that for a Hermitian preveing maps φ . Under some condition, one can directly compute the distribution of $\varphi \otimes \text{id}_d(X_d)$ under the asymptotic sense. [2]

Definition 3. A linear Hermitian preveing maps $\varphi : M_m \rightarrow M_n$ is said to satisfy UC condition if $(\text{id}_n \otimes \text{tr}_m)(P_t) = d_t I$ where $C_\varphi = \sum_{t=1}^r \rho_t P_t$

Corollary 9. Consider the Hermitian preserving maps φ with representation $C_\varphi = \sum_{t=1}^r \rho_t P_t$ and $\mathcal{SN}(P_t) \leq k$. Then, the followings are equivalent.

1. φ is k -positive.
2. φ is completely positive.
3. There exist a $\tilde{C}_\varphi \in M_n \otimes M_k$ such that $\tilde{C}_\varphi \geq 0$ with $\tilde{C}_\varphi = \sum_{t=1}^r \rho_t \tilde{P}_t$

Proof. We will show that $(1) \Rightarrow (3) \Rightarrow (2) \Rightarrow (1)$.

For $\mathcal{SN}(P_t) \leq k$, according to Lemma 5, there is a $\tilde{P}_t \in M_n \otimes M_k$ such that $\text{id}_n \otimes i(\tilde{P}_t) = P_t$, then, define $\tilde{C}_\varphi = \sum_{t=1}^r \rho_t \tilde{P}_t$. It follows that $\langle \tilde{C}_\varphi, (\text{id} \otimes i)(xx^*) \rangle = \langle C_\varphi, vv^* \rangle \geq 0$. Thus, $\tilde{C}_\varphi \geq 0$.

Now, suppose $\tilde{C} \geq 0$. It admits some one rank decomposition $\tilde{C} = \sum_i \lambda_i x_i x_i^*$. Then, consider $C_\varphi = \text{id}_n \otimes i(\tilde{C}) = \sum_i \lambda_i v_i v_i^*$ for some $\mathcal{SN}(v_i v_i^*) \leq k$. Then,

$C_\varphi \geq 0$, thus the corresponding φ is completely positive, which in turn implies k -positivity. $\square$

Corollary 10. Consider the Hermitian preserving maps φ with representation $C_\varphi = \sum_{t=1}^r \rho_t P_t$.

1. If φ is positive (1-positive) and satisfies UC condition, then we have $\sum_{t=1}^r \rho_t d_t \geq 0$.

Proof. According to Remark 7, we have $(\text{id}_n \otimes B)C_\varphi(\text{id}_n \otimes B)^* \geq 0$ for all $B \in M_{d \times m}$ with $\text{rank}(B) \leq k$.

Then, consider $B_k = h_k e_k^*$, $\sum_{k=1}^n \text{id}_n \otimes \text{tr}_d((\text{id}_n \otimes B_k)C_\varphi(\text{id}_n \otimes B_k)^*) = \sum_{k=1}^n \sum_{i,j=1}^n \text{tr}_d(B_k E_{ij} B_k^*) \varphi(E_{ij}) = \sum_{i=1}^n \varphi(E_{ii}) = \text{id}_n \otimes \text{tr}_n(C_\varphi) \geq 0$.

Next, since φ satisfies UC condition, $\text{id}_n \otimes \text{tr}_m(C_\varphi) = \sum_{t=1}^r \rho_t d_t I_n \geq 0$. Thus, $\sum_{t=1}^r \rho_t d_t \geq 0$. $\square$

Theorem 11. Consider a sequence of bipartite unitarily invariant self adjoint matrices $X_d \in M_n \otimes M_d$ converging to a compactly supported probability measure μ . Let $\varphi : M_m \rightarrow M_n$ be a Hermitian preserving maps satisfying UC condition. Then, $X_d^\varphi := (\varphi \otimes \text{id}_d)(X_d)$ converges to a probability measure μ^φ . Moreover,

$$\mu^\varphi = \boxplus_{t=1}^r \left(D_{\frac{\rho_t}{m}} \mu \right)^{\boxplus m d_t}$$

where $C_\varphi = \sum_{t=1}^r \rho_t P_t$

For a random matrix ensemble whose asymptotic eigenvalue distribution converges to a compactly supported probability measure μ , we can define its asymptotic Schmidt number as

Definition 4.

$$\mathcal{SN}(\mu) = \max\{k : \exists k\text{-positive map s.t. } \text{supp}(\mu^\varphi) \not\subset [0, \infty)\} + 1$$

Toy Models

Some Lemma

Lemma 12.

Reference

Bibliography

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